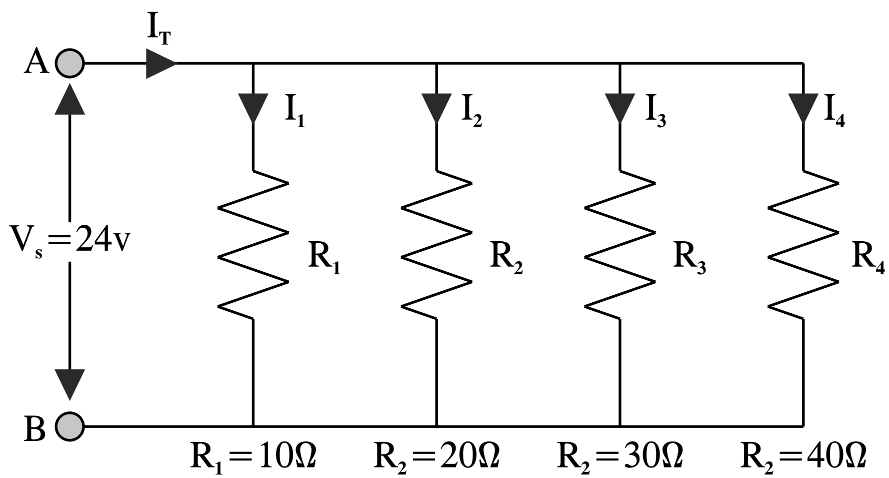


Q.35 What do you mean by a super capacitor?

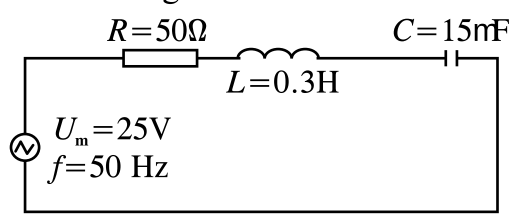
SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Draw the phasor diagram for a series RLC circuit.

Q.37 What are the different types of Semiconductors? What are the different materials used in manufacturing semiconductors?

Q.38 An AC circuit is composed of a serial connection of a resistor with resistance 50Ω, a coil with inductance 0.3 H and a capacitor with capacitance 15 μF. The circuit is connected to an AC voltage source with amplitude 25 V and frequency 50 Hz. Determine the amplitude of electric current in the circuit and a phase difference between the voltage and the current.



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Roll No. ....

4th Sem / Aircraft Maintenance

Subject:- Elements of Electrical and Electronics Engineering-II

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 An electric circuit operates in ________ type of connectivity configuration?

a) Open b) Short

c) Disconnected d) All of the above

Q.2 Which of the following are the examples of electro mechanical component?

a) Bulb b) EMF

c) Motor d) Resistor

Q.3 For a coil having a magnetic circuit of constant reluctance, if the flux increases, what happens to the current?

a) Increases b) Decreases

c) Remains constant d) Becomes zero

Q.4 For a direct current, the rms current is _______ the mean current.

a) Greater than b) Less than

c) Equal to d) Not related to

Q.5 Hysteresis loss in a DC machine depends on ______.

a) Volume and grade of iron

b) Value of flux density

c) Flow rate of ventilating air

d) frequency of magnetic reversal

Q.6 For a direct current, the rms voltage is _______ the mean voltage.
a) Greater than b) Less than
c) Equal to d) Not related to

Q.7 The unit for inductance is _________
a) Ohm b) Henry
c) A/m d) A/s

Q.8 The quality factor Q of a series RLC circuit in terms of inductance L is given by ________
a) w_0L/R b) W_0R/L
c) w_0L/R^2 d) None of the above

Q.9 Which special type of diode is capable of both amplification and oscillation?
a) Point contact diode b) Zener diode
c) Junction diode d) Tunnel diode

Q.10 A thyristor is a:
a) 3 layer, 3 terminal, 3 junction device
b) 4 layer, 3 terminal, 4 junction device
c) 4 layer, 3 terminal, 3 junction device
d) 3 layer, 4 terminal, 3 junction device

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What do you mean by Dielectric loss?
Q.12 Describe Lenz law?
Q.13 What is the function of diode?
Q.14 How do thyristors work?
Q.15 When the voltage across a capacitor increases, what happens to the charge stored in it?
Q.16 What is the material used for resistor?
Q.17 What are variable capacitors?

Q.18 How do inductors work?
Q.19 What is reactive power?
Q.20 How do semiconductors work?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What is the difference between voltage and EMF?
Q.22 Highlight the major difference between AC and DC.
Q.23 Find the relation between a heating coil and water temperature.
Q.24 What is the difference between Apparent Power and Reactive Power?
Q.25 Highlight the major differences between AC and DC.
Q.26 Describe how Kirchoff law is applied.
Q.27 Explain the analogy between electric and magnetic circuits.
Q.28 Explain the working of a photo cell.
Q.29 What are Variable Capacitors? Where are they used?
Q.30 Explain the different types of material used in making various kind of components.
Q.31 Explain the concept of B-H Curve and magnetic Hysteresis.
Q.32 What is asemiconductor? Give examples.
Q.33 Derive the Equation of an alternating voltage and current and wave shape varying sinusoidally.
Q.34 Calculate the individual branch currents and total current drawn from the power supply for the following set of resistors connected together in a parallel combination.